

Parth Sarthi Mishra

P2P Systems Engineer | Distributed Systems | Rust

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EXPERIENCE

ZOOPER

Backend Intern

On-Site

April 2026 - May 2026

- Engineered and deployed a **high-availability API** for real-time driver device-health monitoring and permission tracking.
- Built **telemetry monitoring** and validation workflows, then **stress-tested** the system to identify and fix issues causing data loss and delayed status updates.
- Developed Playwright e2e tests and integrated **MSW** to **Mock API** responses and validate workflows without relying on backend services.

SYSTEM & PROTOCOL PROJECTS

🔗 [CORROBORATE - Distributed State Sync Library](#) (Ongoing)

[GitHub](#)

Rust, CRDTs, Distributed Systems, Synchronization

- Building a Rust library to maintain and synchronize shared state across independent replicas in a p2p environment.
- Designing it to tolerate distributed-systems conditions such as **duplicate, delayed, reordered, and temporarily unavailable messages**.
- Developing tests for **duplicate, delayed, and reordered operations** to verify synchronization and convergence.

🔗 [CHORRENT - P2P File Transfer Library](#)

[GitHub](#)

Rust, P2P Networking, Concurrent File Transfer

- Built a Rust-based P2P file-transfer library from scratch for transferring files between independent peers.
- Designed the transfer system around **peer communication, pieces/chunk-based data** movement, and concurrent transfer workflows.
- Focused on understanding practical P2P concerns including **data integrity, concurrency, and failure handling**.

🔗 [ECHOIT MESSENGER - PEER-TO-PEER COMMUNICATION](#)

[GitHub](#)

Rust, WebRTC, P2p, Documentation

- Designed a decentralized P2P messaging system for peer communication and data exchange without relying on a centralized source of truth.
- Authored the project's **PRD, RFCs, system architecture, and protocol requirements**, defining component boundaries and communication flows across the P2P stack.

🔗 [DICSUSSION PROTOCOL STACK](#)

[GitHub](#)

Protocol Design, RFCs, Distributed Systems

- Designed the communication protocol and defined its message flow, system requirements, and protocol boundaries.
- Authored RFCs and technical specifications used as the communication foundation for EchoIt.

OPEN SOURCE CONTRIBUTIONS

CODEFIT — Open Source Contributor

[GitHub](#)

- Contributed to an open-source interview platform through multiple merged PRs, working on login, email verification, interviews, and notifications.
- Added **Judge0 code execution** to the backend and connected it to the WebSocket system so interview participants could receive code results in real time.
- Reported and helped track bugs related to camera controls and the programming-language selector through GitHub issues.

SKILLS

Languages: Rust, C++, Go, Python, Java

Systems & Networking: P2P Networking, Distributed Systems, Protocol Design, Concurrency, WebRTC, TCP/IP Fundamentals

Tools: Git, GitHub, Docker, SQLite, Linux